

PROJECT EXPO(2026)

PIEZOELECTRIC FOOTSTEPS POWER GENERATING SYSTEM

TEAM MEMBERS:

1. DHANUSIYA A
2. SHRINITHI K M
3. NADHIYA G

PROBLEM STATEMENT:

Large amounts of energy produced by human footsteps are wasted everyday in crowded places such as railway stations, shopping malls, airports, and colleges. At the same time, the demand for electricity is continuously increasing, while conventional energy sources are limited and cause environmental pollution. There is a need for a clean, renewable, and sustainable method of generating electricity. This project addresses the problem by converting the mechanical energy from footsteps into electrical energy using piezoelectric sensors, which can be stored and used for low-power applications.

ABSTRACT :

The piezoelectric footstep power generating system is an innovative renewable energy project that converts the mechanical energy produced by human footsteps into electrical energy using piezoelectric sensors. As the demand for electricity continues to increase and conventional energy resources become limited, there is a growing need for alternative, eco-friendly power generation methods. Every day, a large amount of mechanical energy is wasted in crowded places such as railway stations, bus terminals, shopping malls, airports, schools, and colleges. This project aims to utilize that wasted energy effectively.

The working principle is based on the piezoelectric effect, in which piezoelectric materials generate an electric voltage when mechanical pressure is applied. When a person walks over the footstep platform, the pressure exerted on the piezoelectric sensors produces electrical energy. The generated alternating current (AC) is converted into direct current (DC) using a rectifier circuit and then stored in a rechargeable battery. The stored energy can be used to power LED lights, digital displays, sensors, or other low-power electronic devices.

This system is environmentally friendly, cost-effective, and requires minimal maintenance. It reduces dependence on conventional electricity sources while promoting the use of renewable energy. The project also raises awareness about energy conversion by demonstrating how everyday human activities can contribute to power generation. Although the amount of electricity generated from a single step is small, installing multiple piezoelectric sensors in high-traffic areas can significantly increase the overall energy output.

In conclusion, the piezoelectric footstep power generating system provides a practical and sustainable solution for harvesting wasted mechanical energy. It has great potential for smart cities and public infrastructure, where it can contribute to clean energy production and support the development of a greener and more energy-efficient future.

OBJECTIVES:

- ❖ To design and develop a compact footstep-based energy harvesting system using piezoelectric sensors.
- ❖ To study the performance and efficiency of piezoelectric materials under different pressure conditions.
- ❖ To optimize the arrangement of piezoelectric sensors for maximum energy output.
- ❖ To develop a reliable circuit for collecting, regulating, and storing the generated electrical energy.
- ❖ To demonstrate a low-cost and scalable solution suitable for public infrastructure projects.
- ❖ To encourage the adoption of innovative technologies that utilize otherwise unused mechanical energy.
- ❖ To evaluate the feasibility of integrating the system into smart buildings and smart city applications.
- ❖ To promote practical learning about renewable energy and energy harvesting technologies through a working prototype.

SCOPE OF THE PROJECT:

- ❖ **Footstep energy harvesting:** Designing and developing a floor tile prototype embedded with piezoelectric transducers to capture kinetic energy from human footsteps.
- ❖ **Power conditioning circuit:** Implementing rectifiers, buck-boost converters, and voltage regulators to convert AC spikes from crystals into stable DC power.
- ❖ **Energy storage integration:** Incorporating a rechargeable battery or supercapacitor bank to store the regulated electrical energy for later use.

- ❖ **Target installation environments:** Focusing deployment scope on high-foot-traffic indoor areas such as railway stations, subway corridors, malls, and university walkways.
- ❖ **Real-time monitoring system:** Integrating a microcontroller and LCD or IOT module to track and display generated voltage, current, and total power in real time.
- ❖ **Material evaluation:** Restricting transducer selection to commercially viable and high-yield piezoelectric materials like Lead Zirconate Titanate (PZT) or Polyvinylidene Fluoride (PVDF).
- ❖ **Low-power application load:** Powering small-scale, localized systems such as emergency LED lighting, indicator signs, or low-voltage mobile charging ports.
- ❖ **Efficiency and durability optimization:** Analysing structural wear and electrical degradation under continuous impact to ensure a prolonged operation lifespan.
- ❖ **Cost and scalability analysis:** Evaluating the manufacturing costs of individual tiles to determine the financial feasibility of large-scale grid deployment

EXPECTED OUTCOMES:

- ❖ **Functional prototypes:**
Successful fabrication of a durable, responsive energy-harvesting floor tile that generates electricity upon mechanical deformation.
- ❖ **Stable power output:**
Delivery of a steady, rectified, and regulated DC voltage output suitable for powering standard low-consumption electronic devices.
- ❖ **Measurable energy storage:**
Demonstration of efficient energy accumulation within a battery or supercapacitor over a fixed period of simulated foot traffic.
- ❖ **Accurate system telemetry:**
Real-time data visualization of power generation metrics via an onboard display dashboard or an IOT application interface.
- ❖ **Optimized transducer configuration:**
Identification of the most effective structural layout (series vs. parallel connection) of piezoelectric sensors for maximum power yield.
- ❖ **Quantifiable power metrics:**
Accurate generation of data showing the exact milliwatts or millijoules produced per single standard human step.
- ❖ **Mechanical structural integrity:**
Verification that the tile housing can withstand repeated high-pressure loads without structural deformation or sensor cracking.
- ❖ **Self-sustainable micro-grid blueprint:**

A scalable model showing how multiple tiles can be linked together to autonomously run a localized lighting zone

❖ **Comprehensive cost-per-watt analysis:**

An explicit breakdown of protection costs against energy savings to calculate the estimated return on investment (ROI) timeframe.

❖ **Validated technical documentation:**

Complete schematic diagrams, simulation models and structural designs prepared for subsequent manufacturing phases.

Literature survey:

S.No	Author(s)&Year	Research Focus	Methodology	Research Gap/Contribution
1	Adeel Ali et al.(2021)	Development of a piezoelectric footstep energy harvesting system	The researchers employed multiple piezoelectric discs connected in series with a bridge rectifier and rechargeable battery to convert to mechanical pressure into electrical energy	The study proved the feasibility of harvesting energy from footsteps but produced limited power output because of a small area.
2	Bingqi Zhao et al.(2023)	Comparative analysis of piezoelectric energy harvester structures	Various harvester configuration, including cantilever, curved beam, flat plate and flex tensional mechanisms, were analysed to evaluate their electrical performance.	The review concluded that structural optimization significantly improves power generation and device durability, providing design guidelines for future systems
3	K.R Hanumantha Rao et al.(2024)	Sustainable energy generation from public pedestrian movement	A prototype floor tile embedded with piezoelectric sensors was designed and tested under different loading conditions to evaluate voltage generation.	The work demonstrated the suitability of footstep power generation for public infrastructures such as railway

				stations shopping malls and educational institutions.
4	Samuel oluwaseun lawal et al.(2025)	Intelligent piezoelectric energy harvesting system	The proposed model integrated piezoelectric transducers with a microcontroller-based energy management circuit for efficient storage and monitoring of generated power.	The research improved energy utilization through effective power conditioning but recommended further enhancement in long-term reliability and large-scale implementation.
5	Recent Research (2025-2026)	Smart piezoelectric flooring for IOT and smart city applications	Advanced studies combined piezoelectric floor tiles with wireless monitoring, rechargeable storage units and IOT-enabled control systems for real-time energy management.	The integration of smart monitoring and energy storage increased system efficiency and expanded applications in smart building, public lighting and sensor networks.

Existing Solutions Analysis

1. PZT Ceramic Transducer Flooring Tiles

PZT (Lead Zirconate Titanate) ceramic transducers are the most commonly used piezoelectric materials in footstep power generation systems. Multiple PZT discs are connected in series-parallel configurations and placed beneath a pressure plate. When a person steps on the tile, the mechanical pressure deforms the PZT discs, producing an alternating current (AC) voltage through the piezoelectric effect.

Advantages

- High sensitivity to mechanical pressure.
- Simple and compact design.
- Direct conversion of footsteps into electrical energy.
- Easy integration with energy harvesting circuits.

Limitations

- Brittle and fragile under shear or excessive force.
 - Produces relatively low power output per footstep (approximately **200–300 mW peak**).
 - Requires many PZT discs to generate useful power.
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This method uses a spring-loaded floor panel connected to a rack-and-pinion mechanism. When someone steps on the panel, the rack moves downward and rotates the pinion gear, which drives a small DC generator to produce electricity.

Advantages

- Generates higher torque than direct piezoelectric systems.
- Provides smoother energy transfer.
- Suitable for continuous mechanical energy harvesting.

Limitations

- Bulky and heavy mechanical structure.
 - Moving parts experience wear and tear over time.
 - Requires regular maintenance.
 - Higher installation complexity.
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3. Pavegen-Style Electromagnetic/Piezo Hybrid Walkways

Pavegen-style systems combine electromagnetic generators and piezoelectric technology within modular floor tiles. These smart tiles capture the downward movement of footsteps and convert it into electrical energy, which is commonly used to power LED lighting, sensors, and small micro-grid applications in public spaces.

Advantages

- Highly durable and designed for heavy pedestrian traffic.
- Suitable for commercial installations.
- Can support smart city applications such as LED lighting and environmental sensors.
- Reliable for high-footfall locations.

Limitations

- High installation and deployment cost.
 - Economically feasible only in densely populated areas.
 - Lower efficiency in places with low pedestrian traffic.
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4. Rectification and Battery Storage Circuitry

The electricity generated by piezoelectric sensors is alternating current (AC), which cannot directly power most electronic devices. Therefore, a rectification and storage circuit is required. Typically, a **1N4007 bridge rectifier** converts AC to DC, smoothing capacitors reduce voltage fluctuations, and a voltage booster stabilizes the output to **5V or 12V DC**. The generated energy is then stored in rechargeable lithium-ion batteries.

Advantages

- Converts unstable AC output into stable DC power.
- Enables efficient battery charging.
- Provides continuous power supply for electronic devices.
- Improves usability of harvested energy.

Limitations

- Power loss occurs across diode voltage drops.
 - Passive filter components reduce efficiency during low-power inputs.
 - Energy conversion efficiency decreases when footstep frequency is low.
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Gap Analysis:

1. Energy Conversion Efficiency

- The electrical energy generated from each footstep is very low.
- A significant portion of the mechanical energy is lost due to friction and deformation.
- Existing systems have poor energy conversion efficiency.

2. Material Durability

- Piezoelectric materials (such as PZT discs) crack or degrade after repeated foot traffic.
- Continuous loading reduces the lifespan and performance of the sensors.
- More durable and flexible piezoelectric materials are needed.

3. Load Matching and Energy Storage

- The output voltage and current vary with each footstep.
- Bridge rectifiers and voltage boosters introduce additional power losses.
- Efficient energy storage and power management remain a challenge.

4. Scalability and Cost

- Large-scale installation requires many piezoelectric sensors, increasing the overall cost.
- Maintenance and replacement costs are high.
- The energy generated is often insufficient to justify the investment.

5. Power Output Stability

- Electrical output depends on pedestrian traffic.
- During low footfall periods, power generation drops significantly.
- A stable output requires improved energy storage and regulation.

6. Mechanical Design

- Uneven force distribution reduces the effectiveness of the piezoelectric sensors.
- Existing tile designs are often bulky and uncomfortable for users.
- Better structural design is required for maximum force transfer.

7. Environmental Resistance

- Dust, moisture, and temperature changes reduce system performance.
- Proper sealing and weather-resistant materials are essential for outdoor installations.

8. System Reliability

- Sensor failures and wiring issues reduce long-term reliability.
- Existing systems require frequent inspection and maintenance.

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